

Expanded Search for Toroidal Isomers

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Preliminary Exam / Masters Defense

2026-06-30

Liquid Drop Model (1970s) - Original Theory

Spherical Nucleus Binding Energy

$$E_B = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(N-Z)^2}{A} \pm \delta(N, Z)$$

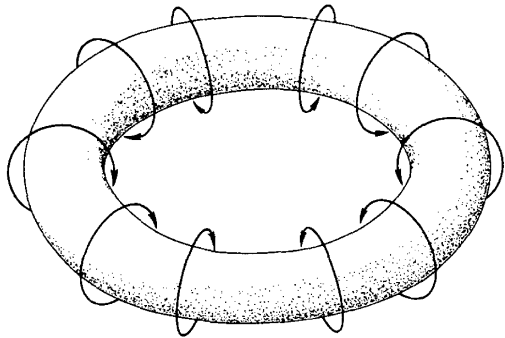
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- E_B depends on **spherical** assumptions.

What happens under deformed Toroidal configurations at high angular momentum?



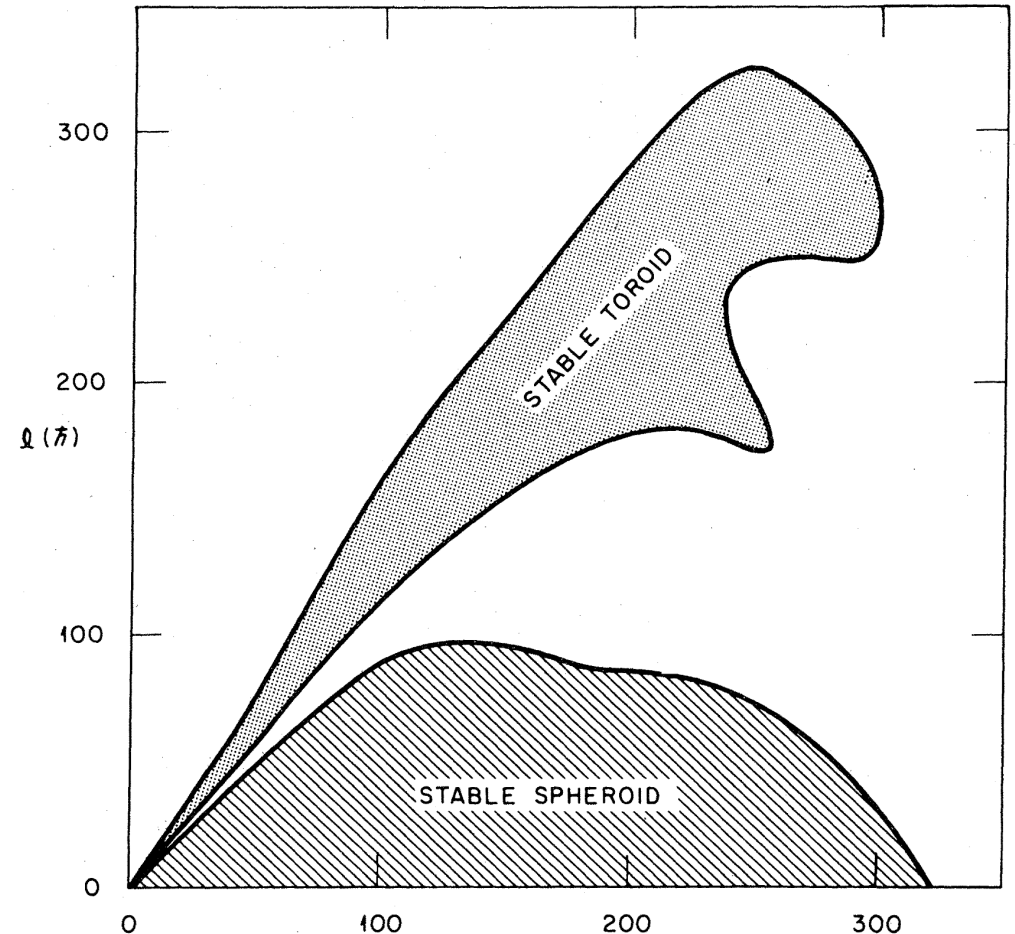
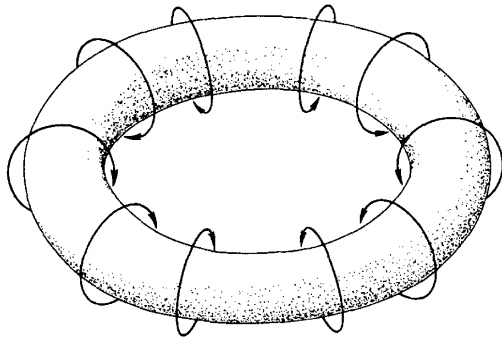
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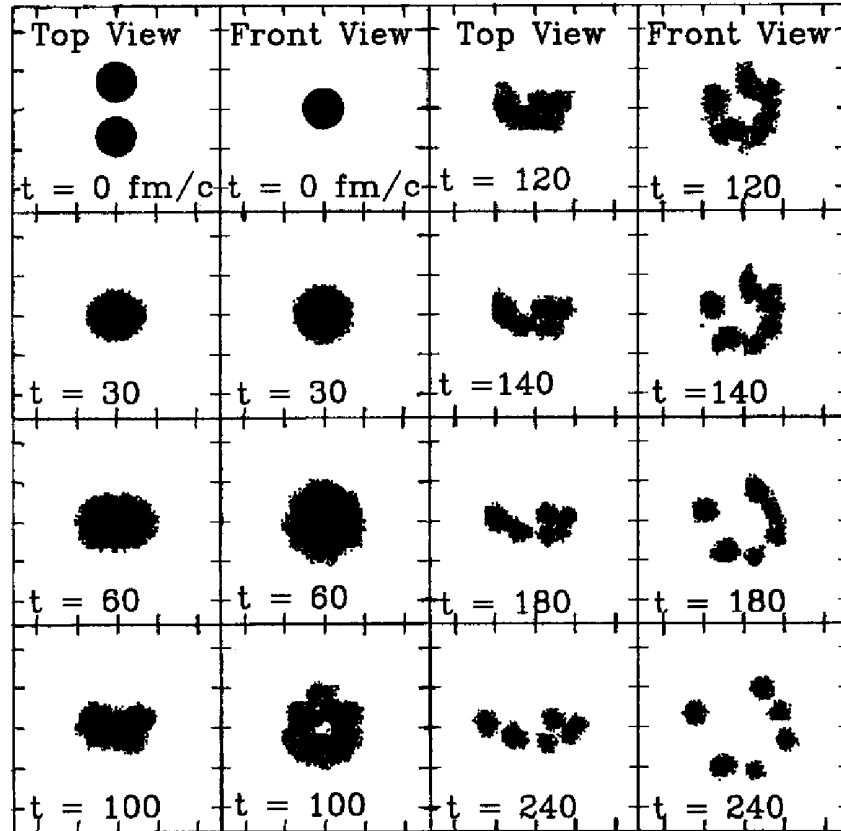
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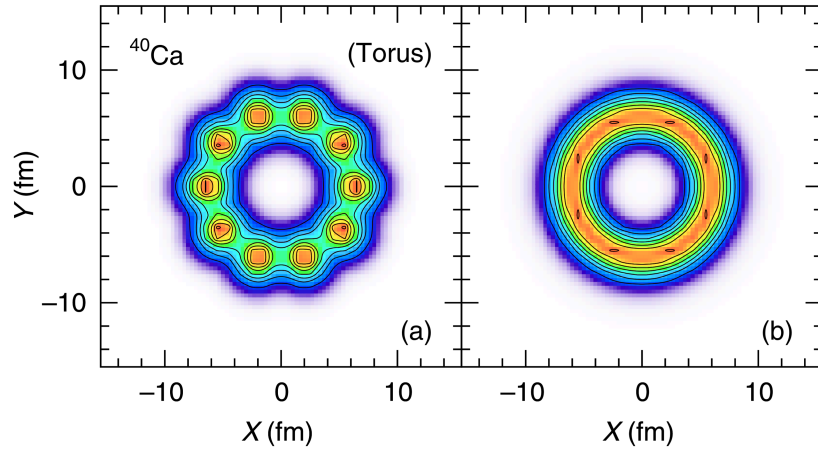


Boltzmann-Uehling-Uhlenbeck (BUU) (1990s) - Dynamic Formation

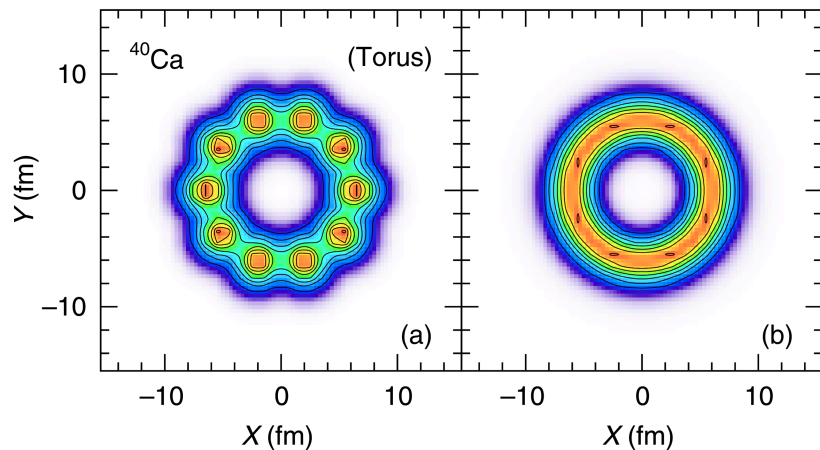
$^{92}\text{Mo} + ^{92}\text{Mo}$, $E/A = 100$ MeV, $b=0$, Stiff EOS



- Direct collisions produced outwardly expanding rings.
- Ring expansions temporarily slowed
- Eventual decay into symmetric particles
 - $r_{\text{particle}} \approx d_{\text{toroid}}$



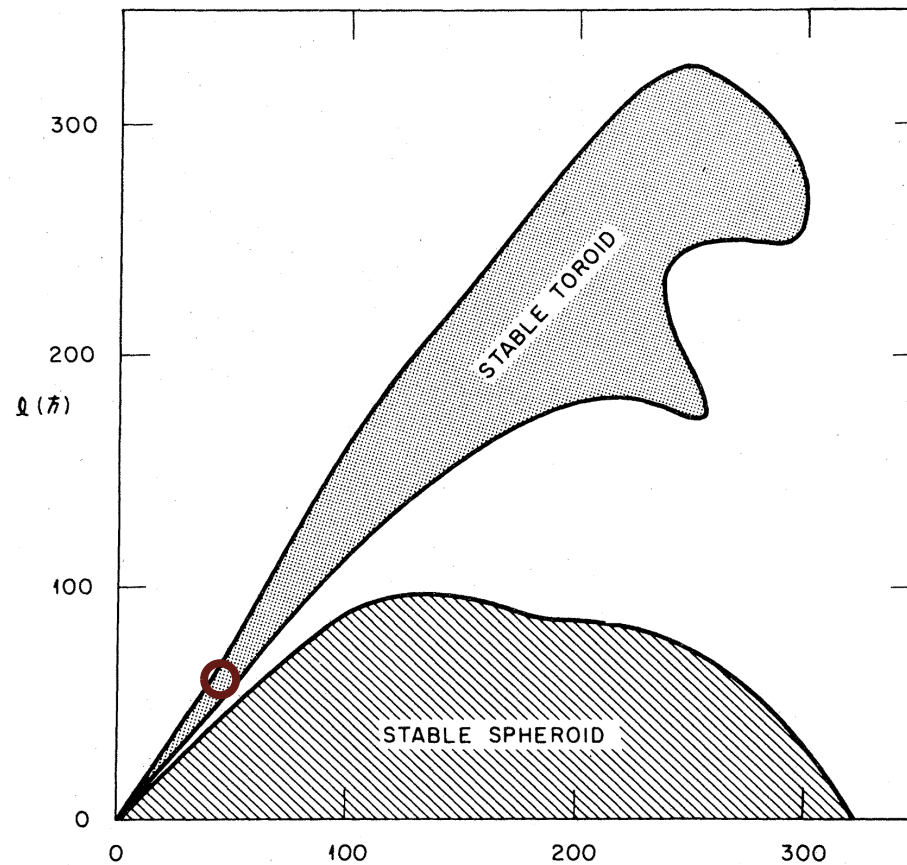
- Predicted state in ^{40}Ca
 - $J_z = 60\hbar$
 - $E^* \approx 175 \text{ MeV}$



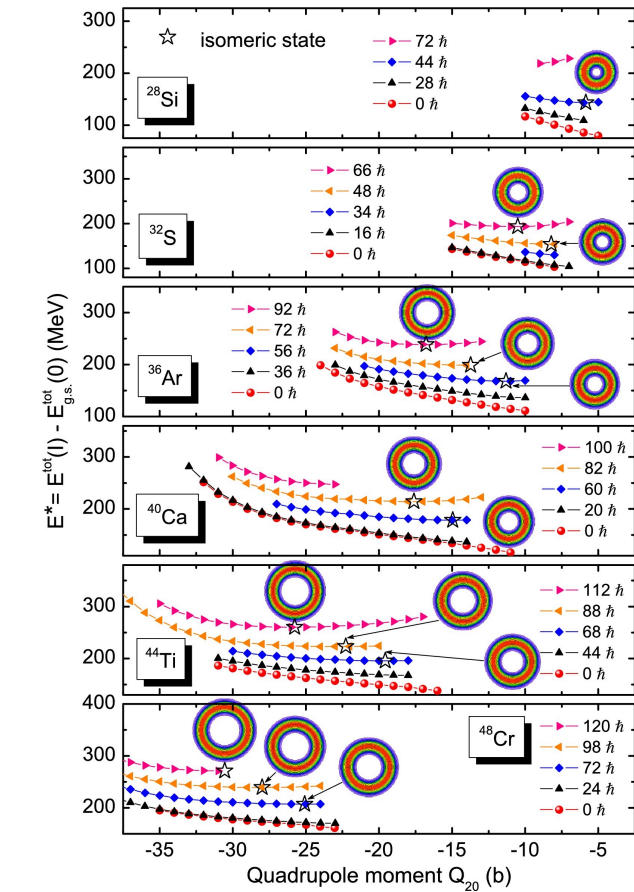
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Angular momentum and mass consistent
with Wong's LDM



Cranked Skyrme Hartree Fock (cSHF) (2010s) - Specific State Predictions



A. Staszczak and C.-Y. Wong predict 18 Toroidal Isomers

	E^* [MeV]	I [$\hbar$]	d [fm]	ρ_{max}/ρ_0
^{28}Si	143.18	44	1.45	0.74
^{32}S	153.87	48	1.42	0.76
	193.35	66	1.40	0.67
^{36}Ar	168.03	56	1.40	0.78
	198.63	72	1.39	0.71
	238.56	92	1.37	0.64
^{40}Ca	178.36	60	1.40	0.79
	214.23	82	1.39	0.73
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Cranked Skyrme Hartree Fock (cSHF) (2010s) - Specific State Predictions

- Ichikawa's state is reproduced
- Many predictions $\Rightarrow$ trends
 - Large low density states
 - Minor axis radius $\sim \alpha$ radius

Toroidal states, should they exist, might decay through several α particles

- Many states to look for

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Minor axis stays about constant ($d \approx 1.4$ fm)

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Low density

Cranked Skyrme Hartree Fock (cSHF) (2010s) - Specific State Predictions

- Ichikawa's state is reproduced
- Many predictions $\Rightarrow$ trends
 - Large low density states
 - Minor axis radius $\sim \alpha$ radius

Toroidal states, should they exist, might decay through several α particles

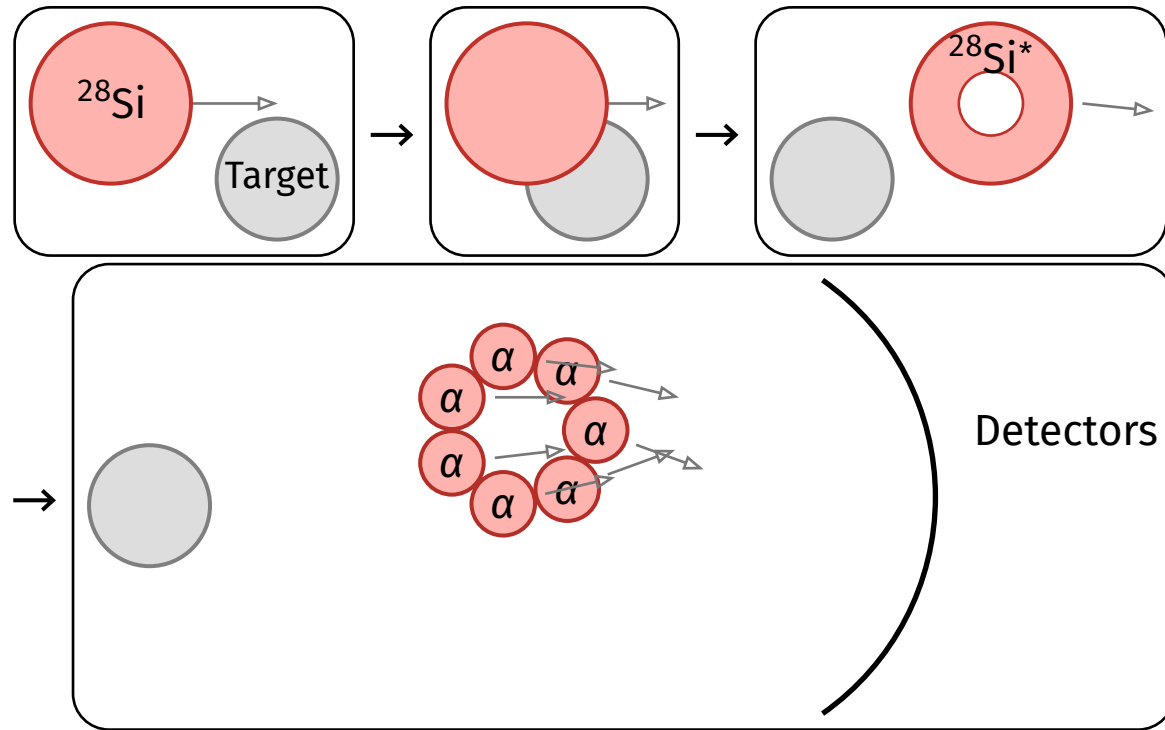
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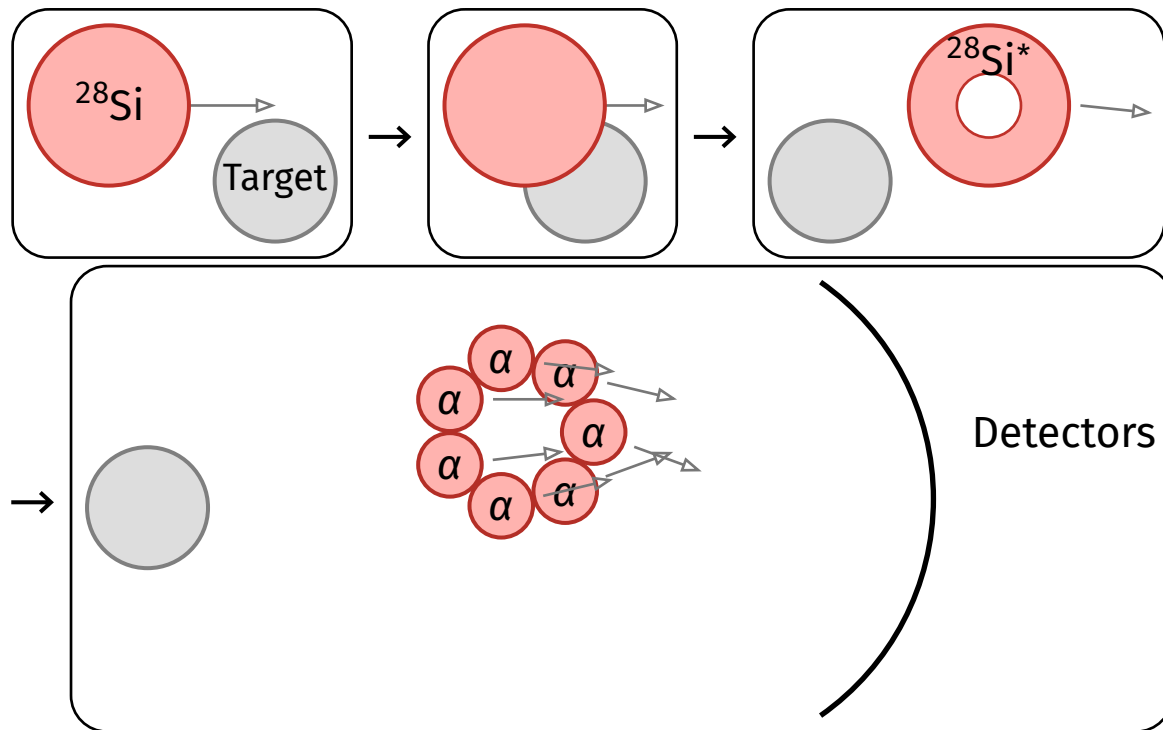
Several states to investigate, including $^{28}\text{Si}(E^* = 143 \text{ MeV})$

First Experimental Evidence (2019)



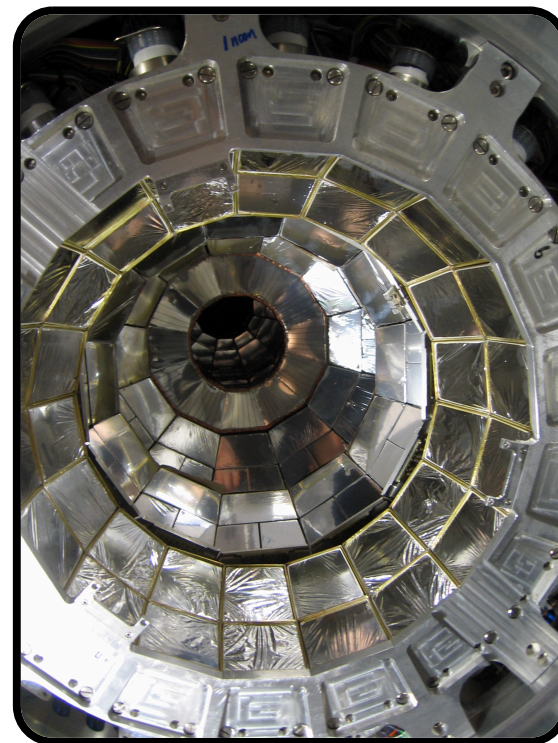
$$E_{^{28}\text{Si}}^* = \underbrace{\sum_i T_{\alpha_i}}_{E_{\text{rel}}} - \underbrace{(m_{^{28}\text{Si}} - 7m_{\alpha})}_Q \quad \text{if } 7\alpha \text{ from } ^{28}\text{Si}^*$$

First Experimental Evidence (2019)



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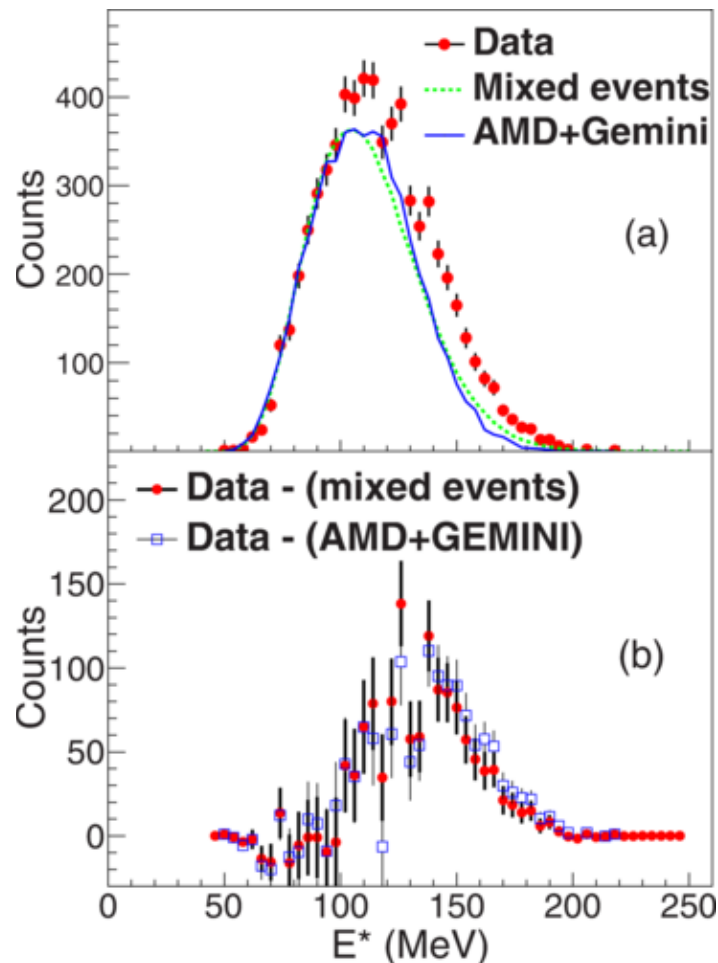
Neutron Ion Multidetector For Reaction Oriented Dynamics (NIMROD)



${}^{28}\text{Si} + {}^{12}\text{C}$ @ 35 MeV/u

First Experimental Evidence (2019) - Cao et al., JBN Group

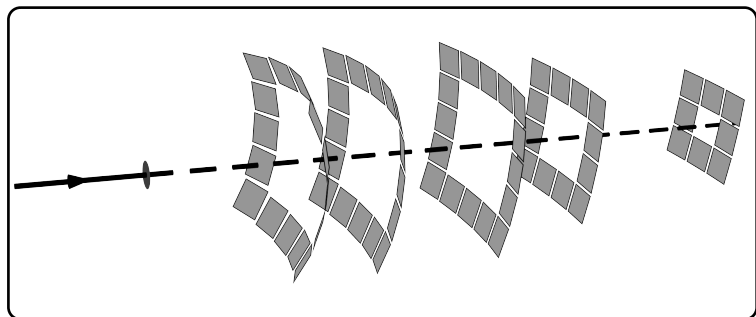
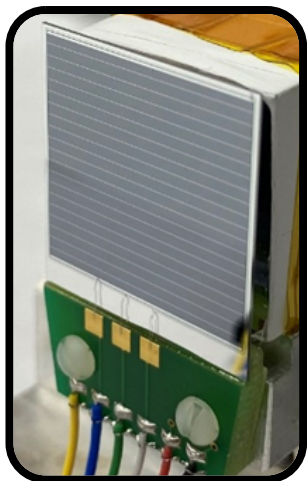
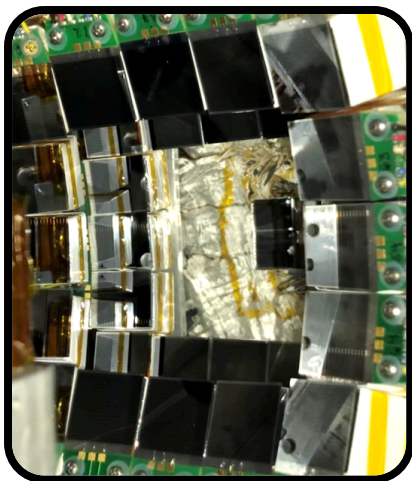
- Reaction
 - $^{28}\text{Si} + ^{12}\text{C}$ @ 35 MeV/u
 - Part of 2009 experimental series
- Sample Size
 - 6,467 Events w/ 7α
- Resolution for E^* from 7α
 - ~ 9.4 MeV (FWHM) in ROI
 - Position insensitive detectors



- Nonresonant Background
 - Mixed Events
 - Shifted AMD distribution
- Extracted Peaks
 - $E^* = 114, 126, \text{ \& } 138$ MeV
 - FWHM dominated by resolution

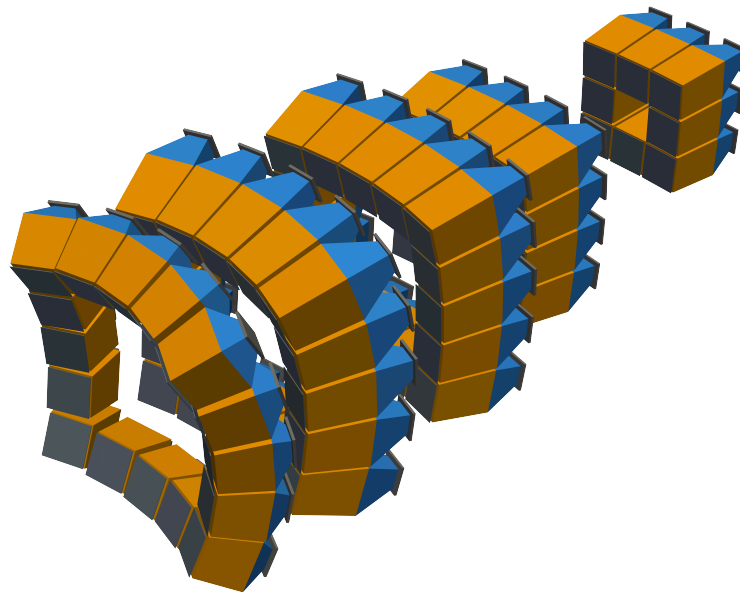
“Clearly an experiment with much better angular resolution, allowing better resolution for the excitation energy spectrum, will be very desirable.”

Forward Array Using Silicon Technology (FAUST)

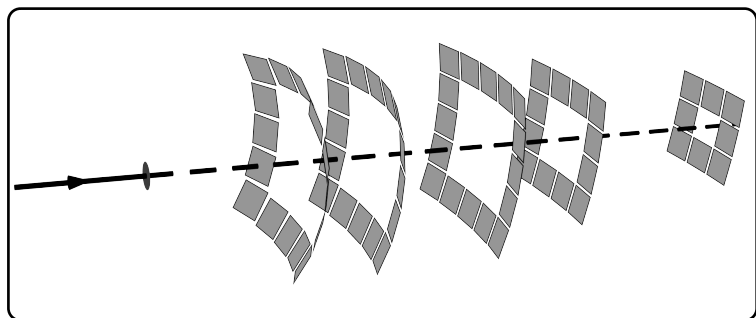
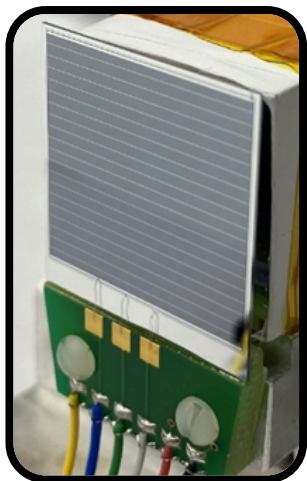
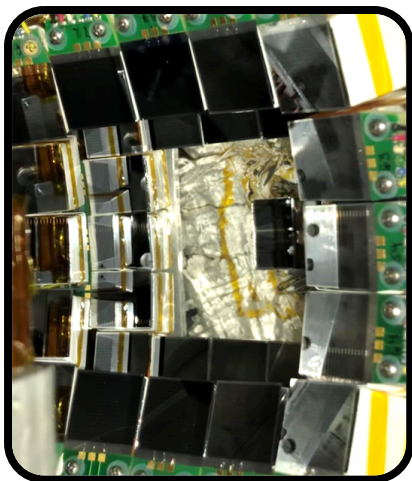


Forward Array Using Silicon Technology (FAUST)

- 68 Si-CsI(Tl) telescopes
 - Excellent Forward Coverage ($1.7^\circ - \sim 40^\circ$)
 - Lab Energy (Thick CsI stops particles)
 - Isotopic Particle Identification ($E-\Delta E$ technique)
 - Lab Angle (Dual-Axis Duo-Lateral detectors)



Forward Array Using Silicon Technology (FAUST)



Forward Array Using Silicon Technology (FAUST)

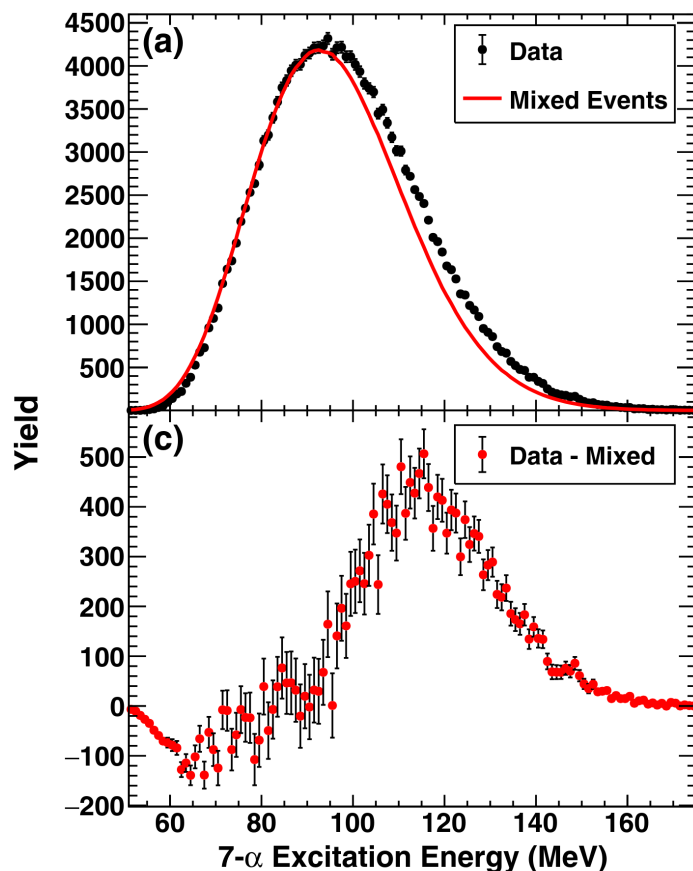
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Measured Reactions

- Target: ^{12}C
- Beam Energy: 35 MeV/u
- Projectiles
 - ^{16}O , ^{20}Ne , ^{24}Mg (Develop analytical techniques + future cluster studies)
 - ^{28}Si (Andy Hannaman's thesis work – confirm and characterize states from Cao *et al*)
 - ^{32}S , ^{36}Ar (Search for new states predicted by Wong)

Second Experimental Result (2023)

- Reaction
 - $^{28}\text{Si} + ^{12}\text{C}$ @ 35 MeV/u
 - Dedicated measurement for toroidal search
- Sample Size
 - 187,067 Events w/ 7α (x29 ↑)
- Resolution for E^* from 7α
 - ~ 2.5 MeV (FWHM) in ROI (x3.5 ↓)



- No obvious peaks in the raw spectrum.
- Need Background
 - Simulations
 - Systematics hard to constrain
 - Mixed Events
 - Known bias (see next slides)

A physically motivated, well benchmarked, and data driven background estimate is required for confident interpretation

One way we know mixed events are imperfect (for this purpose)

A Case Study on d - d - d

Any signals should be **rare to nonexistent**

High energy excitation, deexciting by unlikely channel twice

- $Q(^6\text{Li} \rightarrow 3d) = -25.32 \text{ MeV}$
- $Q(\alpha \rightarrow 2d) = -23.84 \text{ MeV}$
- $Q(d \rightarrow pn) = -2.22 \text{ MeV}$

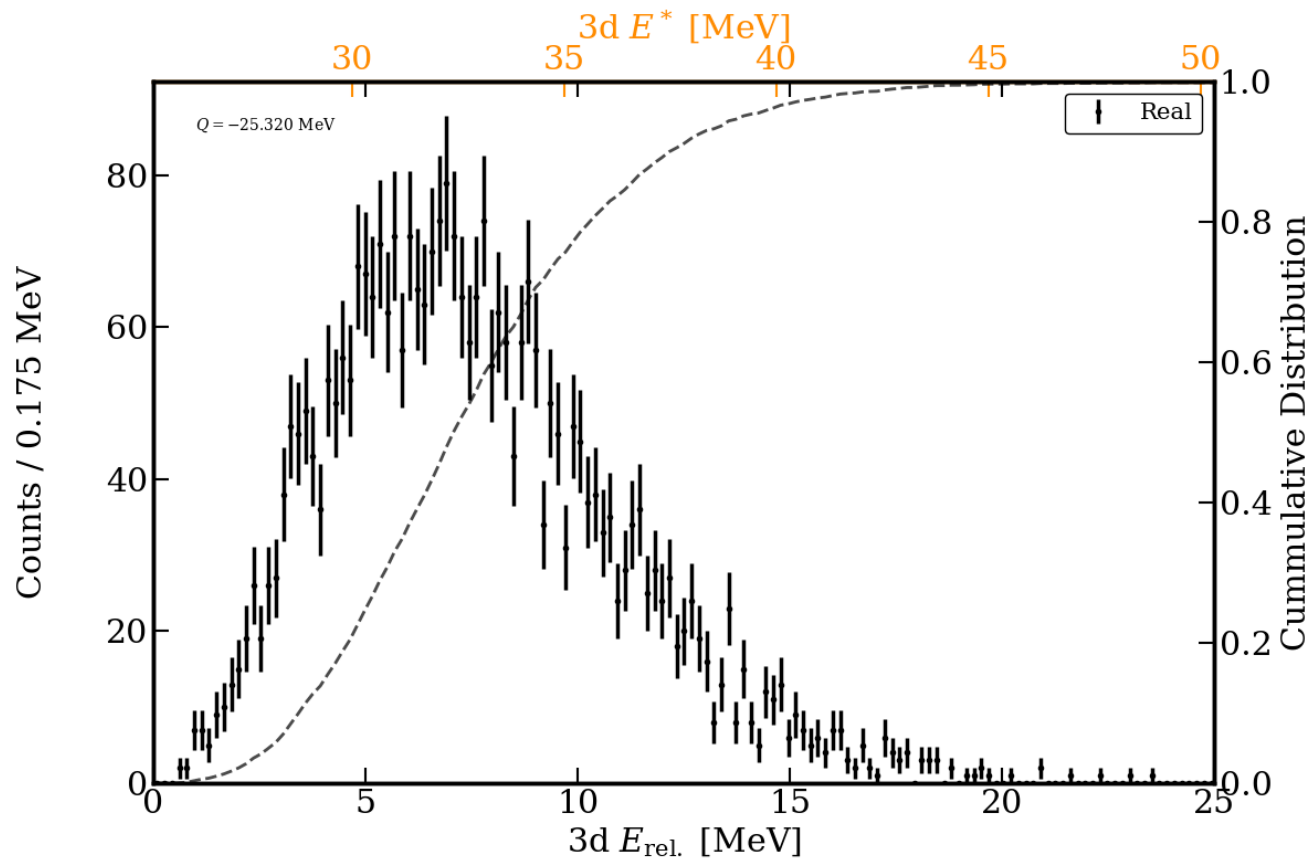
$^6\text{Li}(E^* > 25.32 \text{ MeV})$

$\rightarrow \alpha^*(E^* > 23.8 \text{ MeV})$

+ $d(E^* < 2.2 \text{ MeV})$

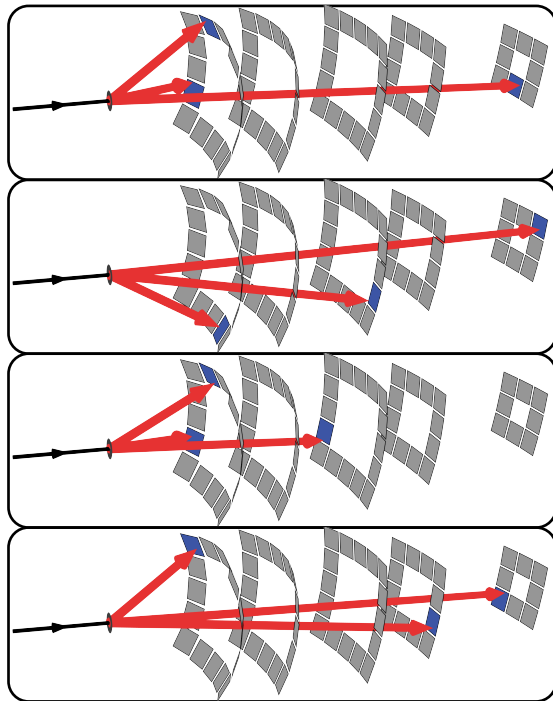
$\rightarrow 3d(E^* < 2.2 \text{ MeV})$

Total = Background + Signal



Creating a Mixed Event

Real Events

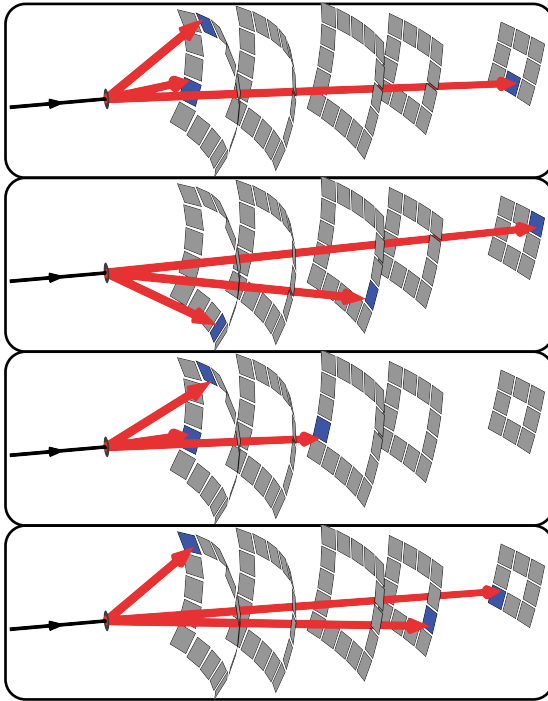


⋮

- Start w/ events containing all particles

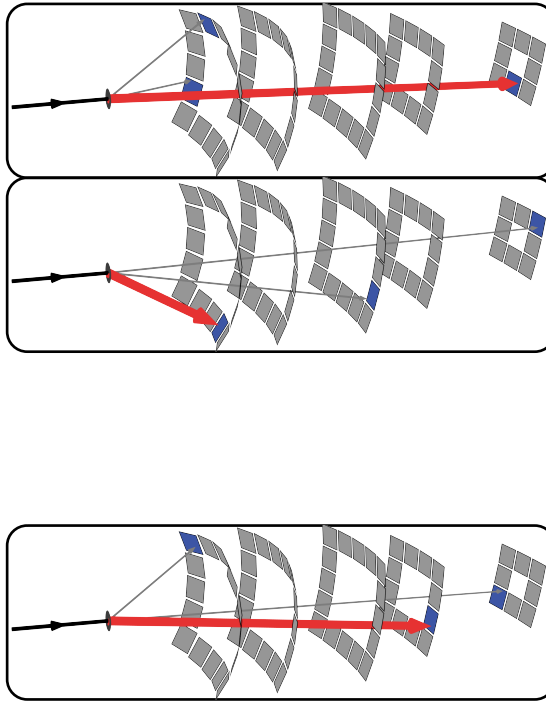
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⋮

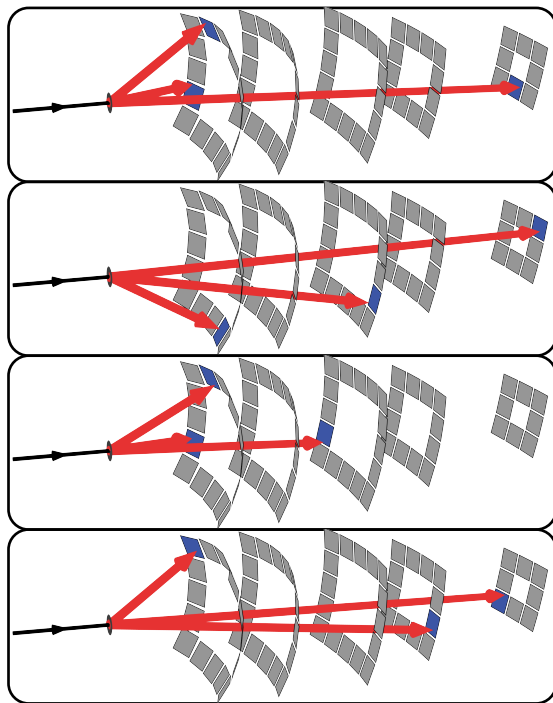
Selected Events



- Start w/ events containing all particles
- Select particles from separate events

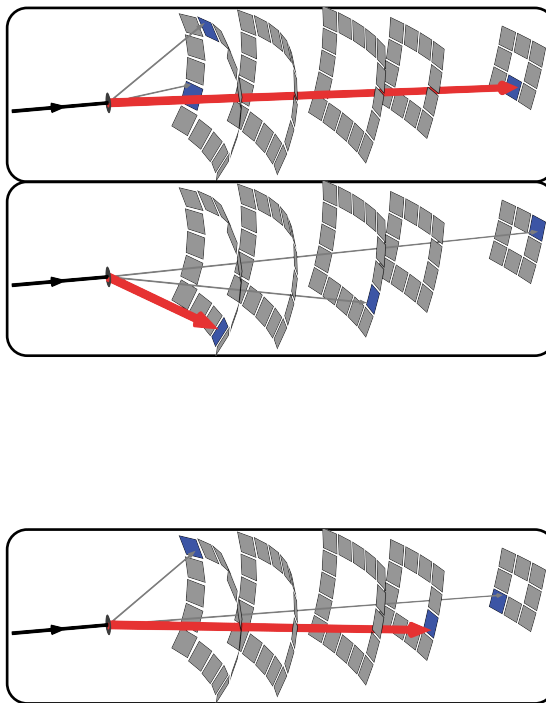
Creating a Mixed Event

Real Events

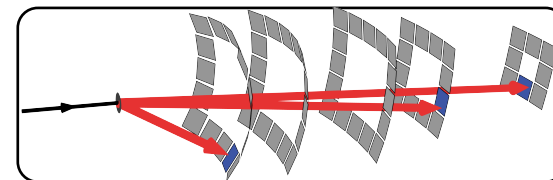


⋮

Selected Events



Mixed Events



⋮

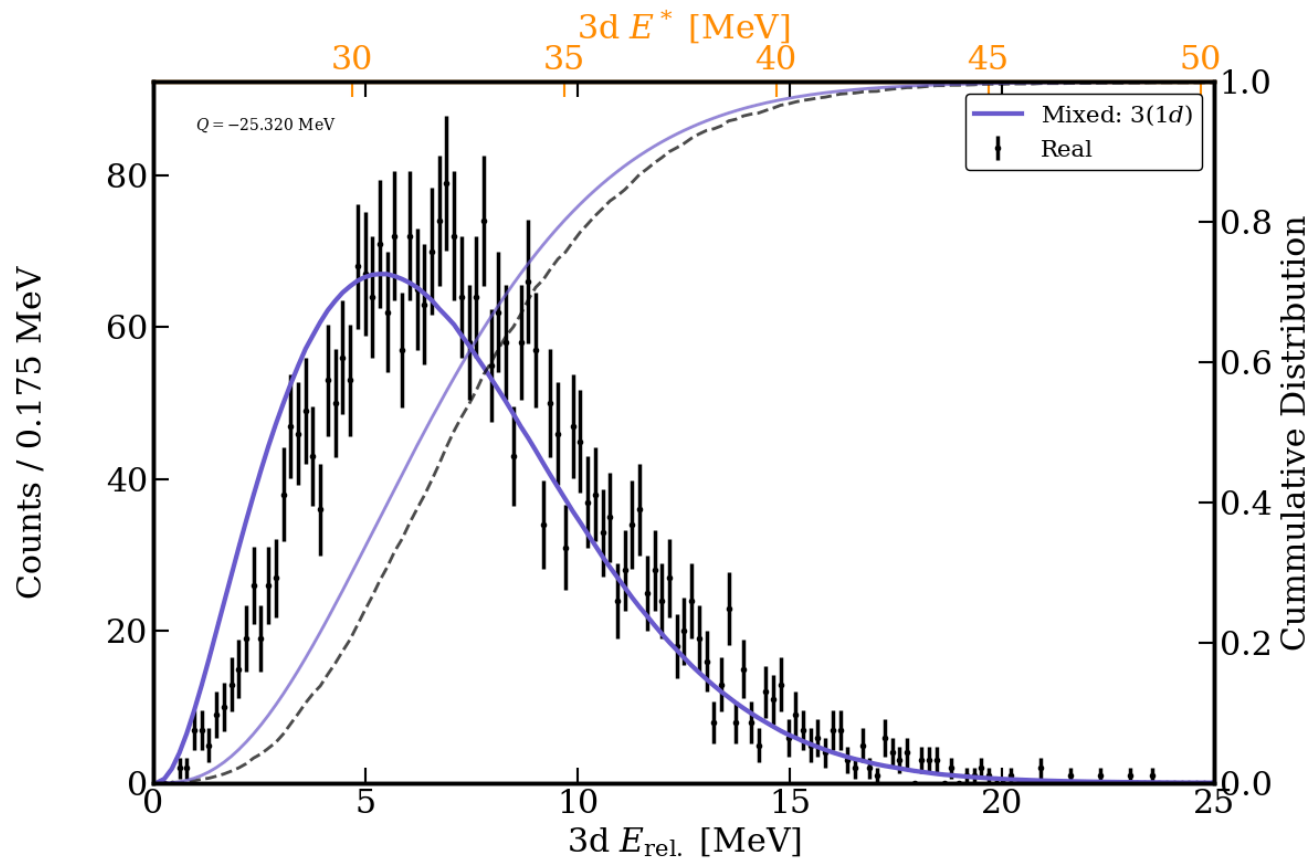
- Start w/ events containing all particles
- Select particles from separate events
- Construct new event

One way we know mixed events are imperfect (for this purpose)

A Case Study on $d-d-d$

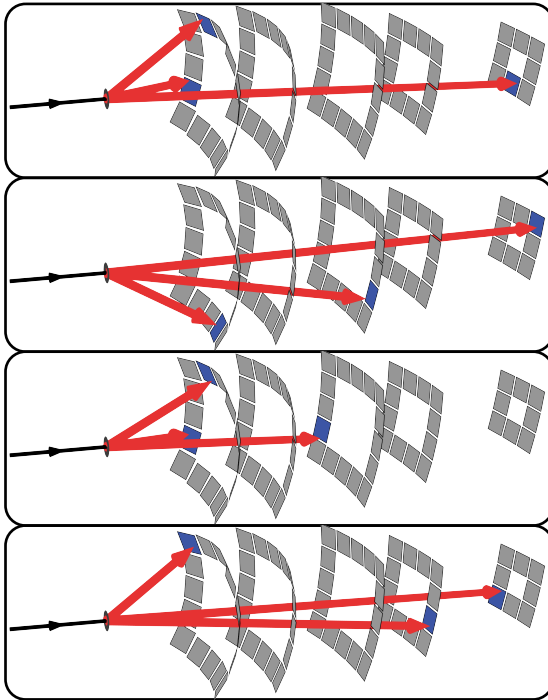
Mixed $\neq$ Total

- Mixed Events removes particle-particle correlations including
 - 3-particle resonances (good)
 - Coulomb repulsion (bad)
 - 2-particle resonances (bad)
- Need a way to **remove 3-particle** correlations while **preserving 2-particle** correlations



Creating a *Partially Mixed Event* [FIXME IMAGES]

Real Events



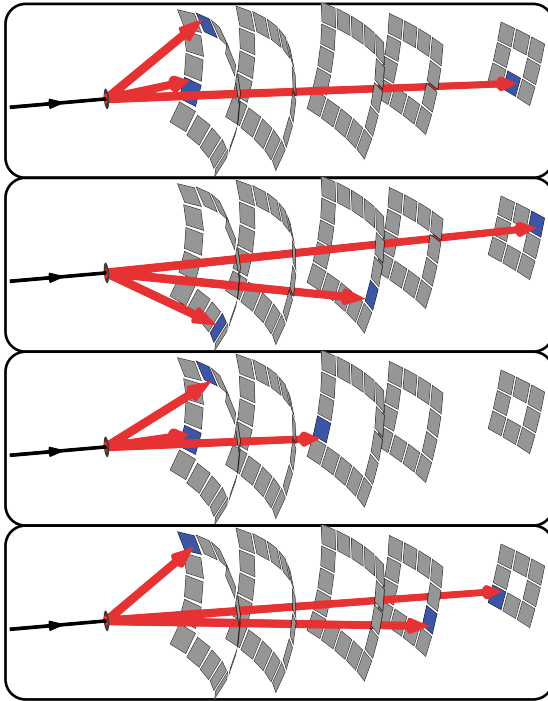
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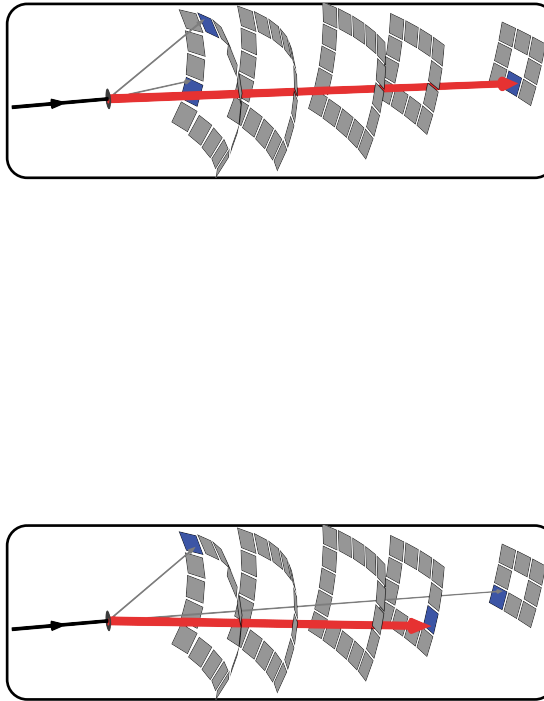
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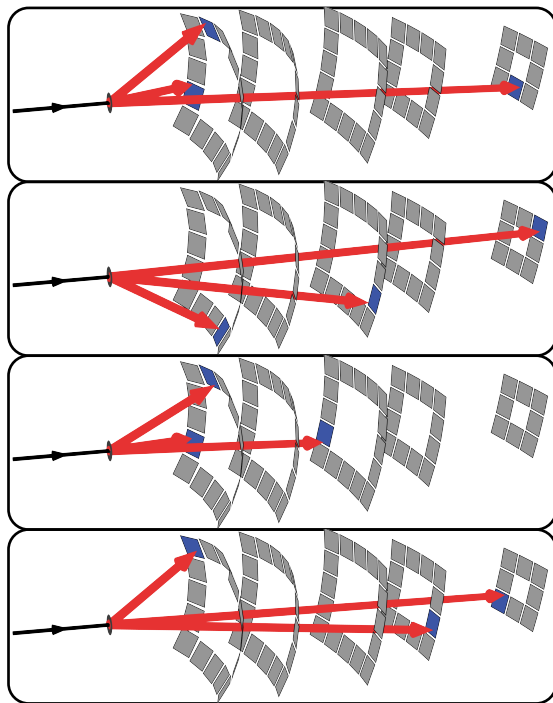
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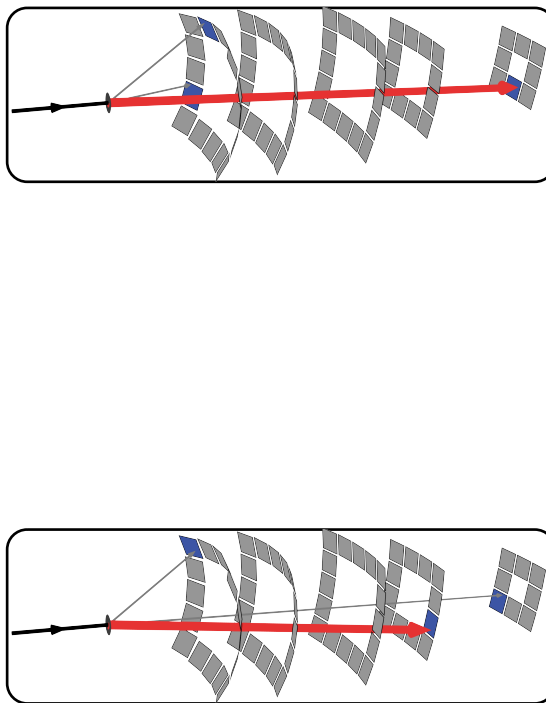
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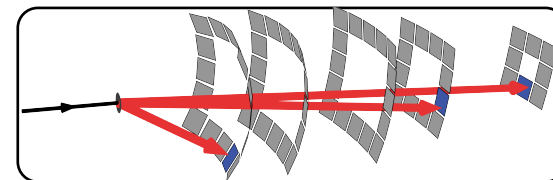


⋮

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Mixed Events



⋮

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Incorporating one of the d - d correlations

A Case Study on d - d - d

- Partial mixing (PM) provides a minor improvement over full mixing (FM).

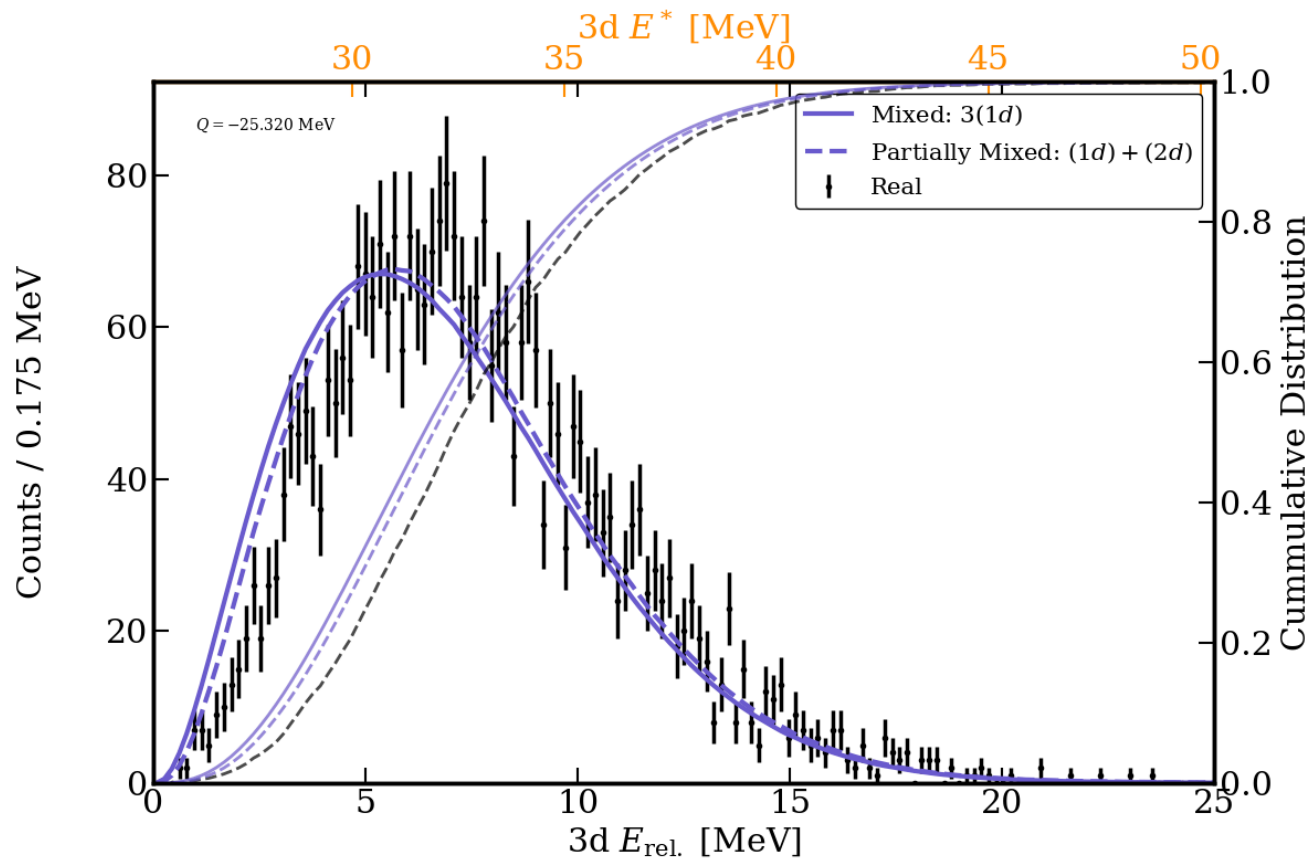
Number of d - d correlations

Real: $\binom{3}{2} = 3$

Partially Mixed: $\binom{2}{2} + \binom{1}{2} = 1$

Mixed: $3\binom{1}{2} = 0$

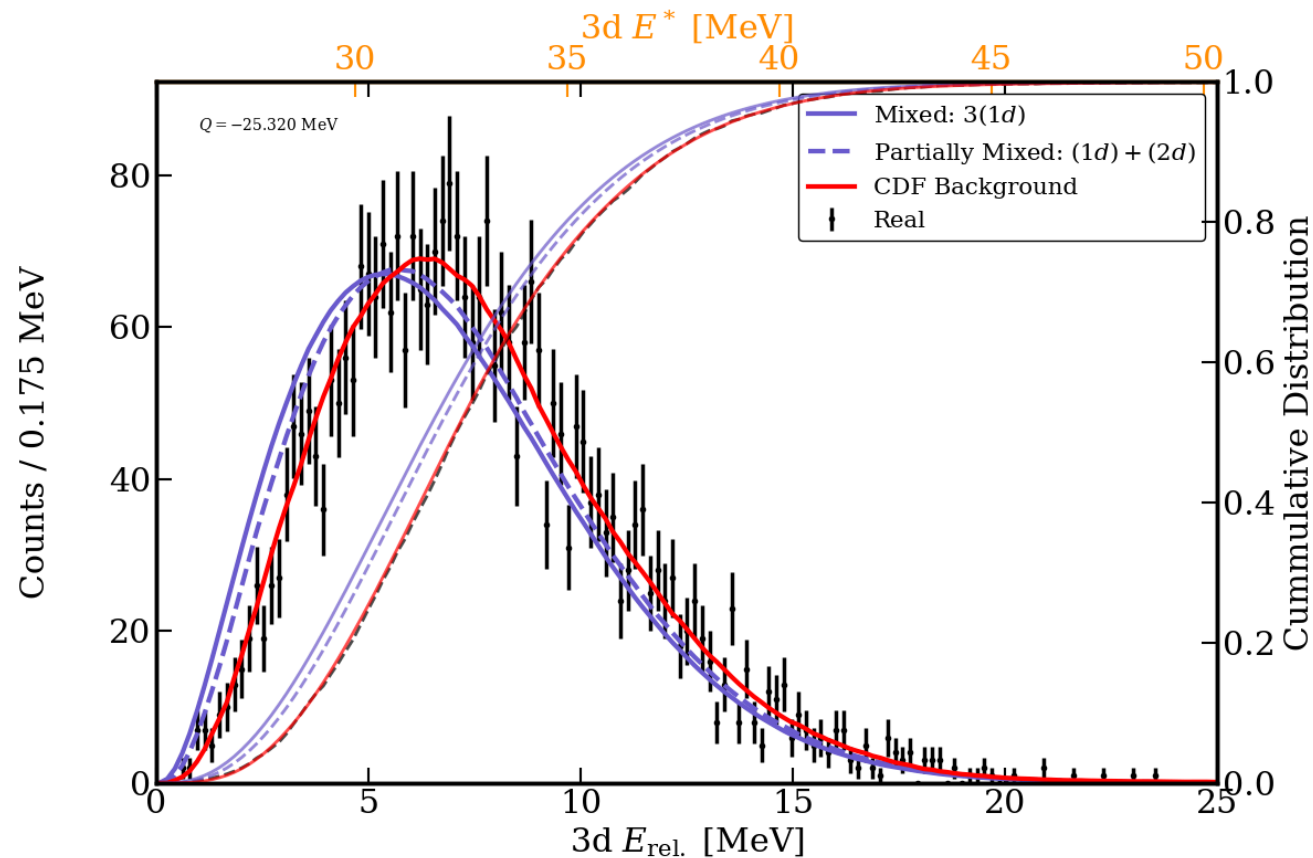
- PM is $\sim \frac{1}{3}$ of the way between FM and Real in the inverse CDF



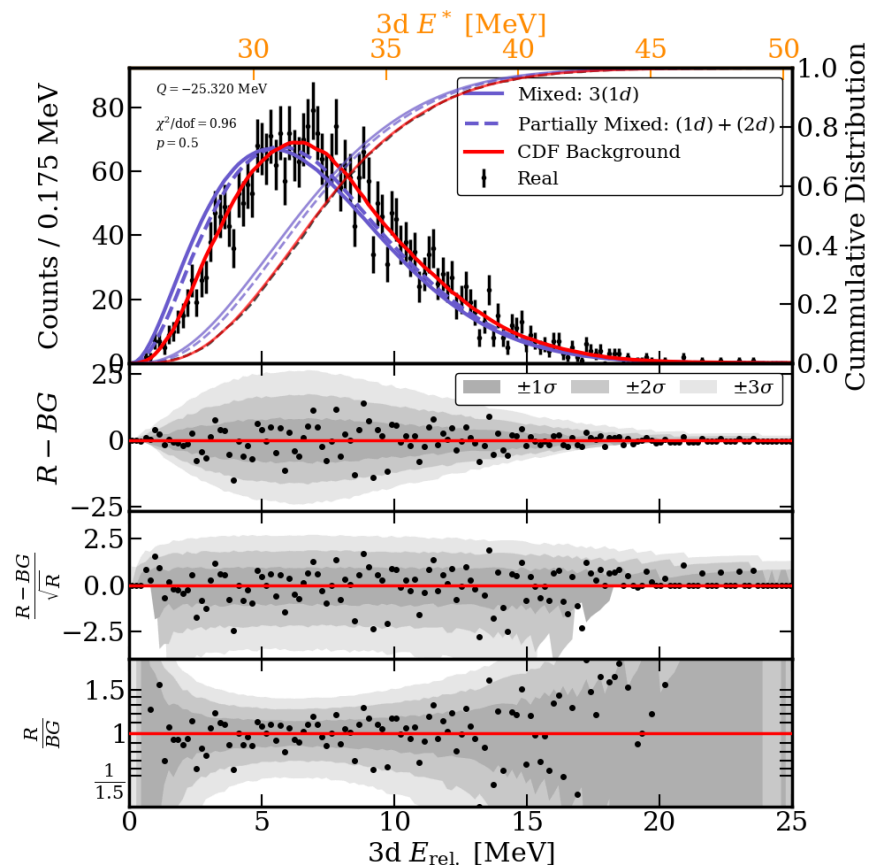
Novel Estimation of the Nonresonant Background

A Case Study on $d-d-d$

1. Measure systematic change by adding a 2-particle correlation through the change from FM to PM.
 2. Propagate that systematic change an additional 2 times to account for all three 2-particle correlations
- Quantile Matching: Horizontal shifts in the CDF
 - Used a lot in climate change studies



Statistical Check



Does the **Background** match the **Total Real**?

- ‘Zero Parameter Fit’
- $\chi^2 / \text{dof} = 0.96$ (1 is ideal)
- $P = 0.4998$ (0.5 is ideal)
 - Resample the model, and assess how often a more extreme χ^2 / dof is observed

The background spectrum is statistically consistent with the measured spectrum.

Real: Allowed to have 3-particle correlations, but physically shouldn't

Background: Constructed in the absence of 3-particle correlations.

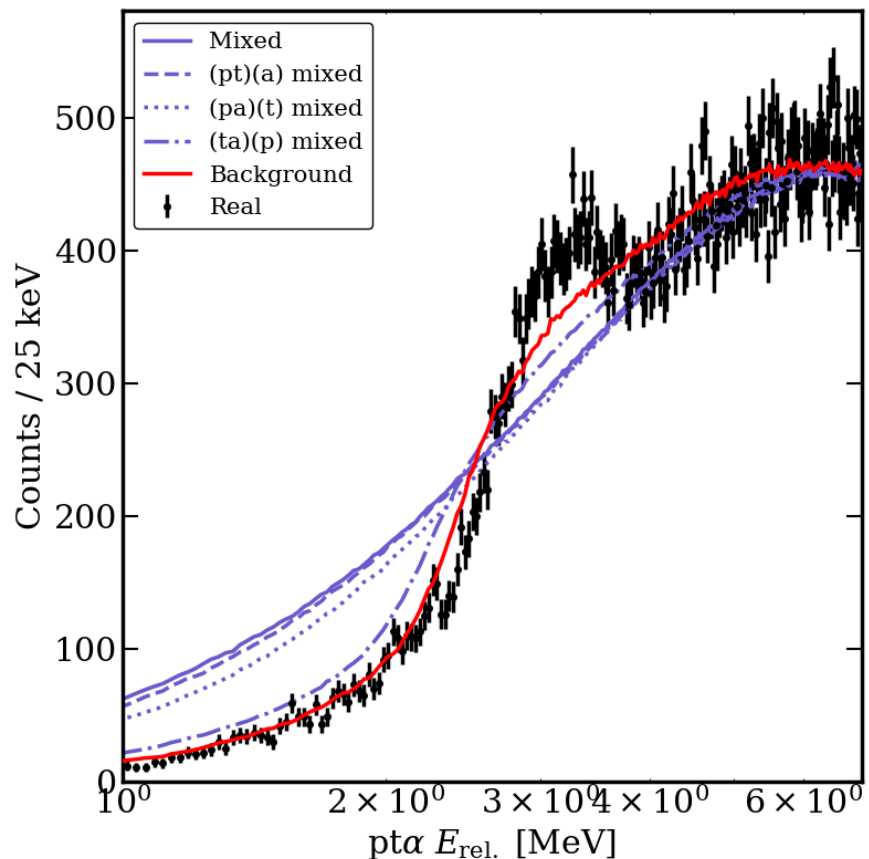
Another Case Study: p - t - α (No Signal)

Does the **BG** match the Total Real?

- $\chi^2 / \text{dof} = 1.23$ (1 is ideal)
- $P = 0.0005$ (0.5 is ideal)

The data is inconsistent with a description of the data that excludes signals.

- A modified model which includes 3-particle correlations is necessary



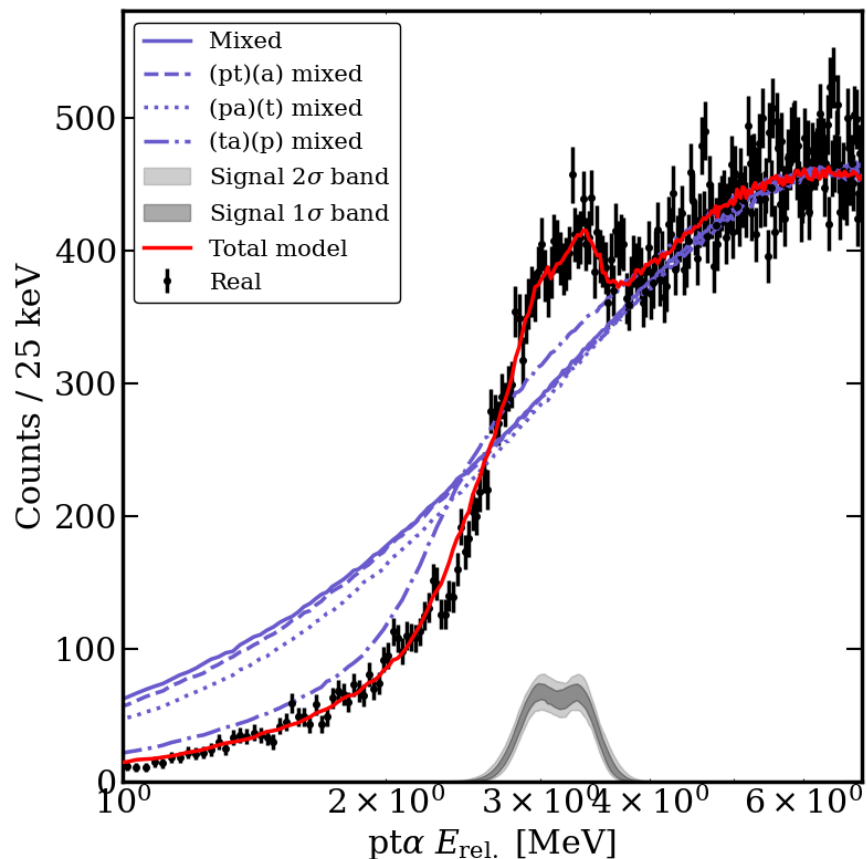
Another Case Study: p - t - α (Crude Double Gaussian)

Very crude (and not fully optimized) double-Gaussian model for signal

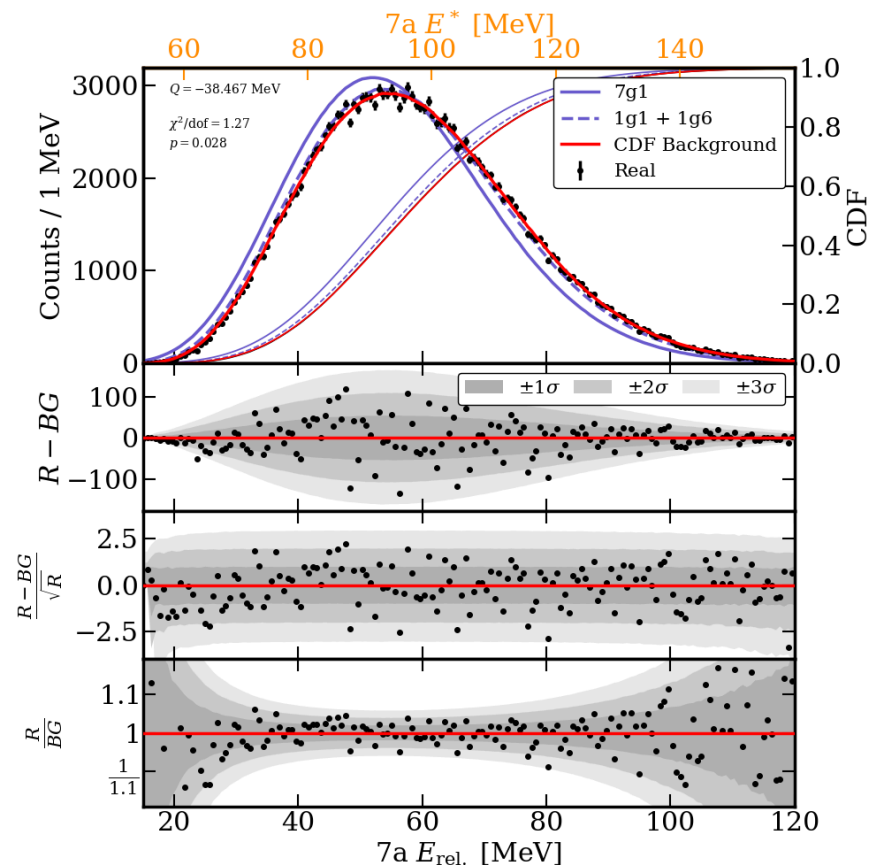
- $\chi^2 / \text{dof} = 1.13$ (1 is ideal)
- $P = [\text{TBD}]$ (0.5 is ideal)

The data is consistent with a description of the data that includes a double Gaussian signal.

Disclaimer: Many details were brushed under the rug here. This fit is actually rather complicated.



7α E* spectrum (no signal model)



Number of α-α correlations

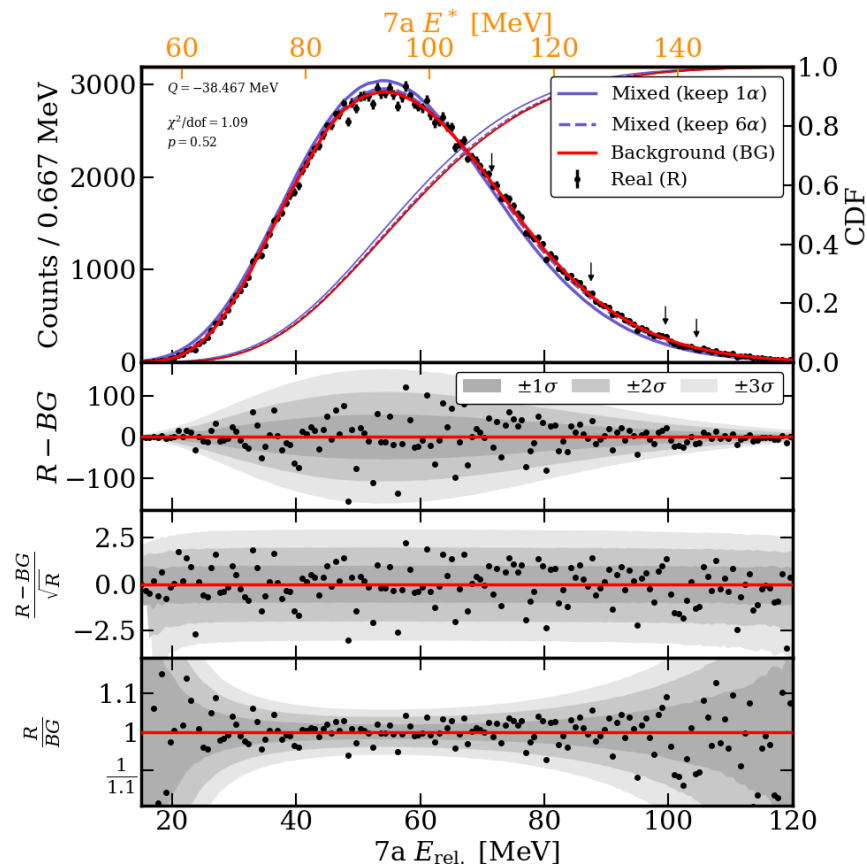
Real: $\binom{7}{2} = 21$

Partially Mixed: $\binom{6}{2} + \binom{1}{2} = 1$

Mixed: $7\binom{1}{2} = 0$

- Partial mixing incorporates 15/21 of the two-particle correlations
- Imperfect, but decent fit
 - $\chi^2 / \text{dof} = 1.15$ (1 is ideal)
 - $P = 0.028$ (0.5 is ideal)
- Minor deviation at low energy
 - Detector hit pattern bias

$7\alpha E^*$ spectrum (no signal model; hit pattern corrected)



- Modify mixing procedure so FM and PM result in same per-event hit pattern as the real data
- Great description of the data
 - $\chi^2 / \text{dof} = 1.09$ (1 is ideal)
 - $P = 0.52$ (0.5 is ideal)
- No perceivable systematic deviations

Conclusions

- Toroidal nuclei are exciting nuclear structures with decades of theoretical support.
- Experimental investigations in 2018 motivated a series higher precision measurements of $N\alpha$ E^* spectra, searching for toroidal isomers in ^{28}Si , ^{32}S , and ^{36}Ar .
- Deeper investigation of the biases of mixed events led to a much more accurate description of the background.
- No evidence for toroidal states was observed in the 7α E^* distribution of $^{28}\text{Si} + ^{12}\text{C}$ @ 35 MeV/u.
 - Upper limits studies are to follow
 - The $N\alpha$ systems of the ^{32}S and ^{36}Ar data are slated to be analyzed shortly

Outline

Contents

Liquid Drop Model (1970s) - Original Theory	2
Spherical Nucleus Binding Energy	2
Boltzmann-Uehling-Uhlenbeck (BUU) (1990s) - Dynamic Formation	5
Cranked Skyrme Hartree Fock (cSHF) (2010s) - Specific State Predictions	8
First Experimental Evidence (2019)	13
First Experimental Evidence (2019) - Cao <i>et al.</i> , JBN Group	15
Forward Array Using Silicon Technology (FAUST)	16
Second Experimental Result (2023)	18
One way we know mixed events are imperfect (for this purpose)	19
A Case Study on d - d - d	19
Creating a Mixed Event	20
One way we know mixed events are imperfect (for this purpose)	23
A Case Study on d - d - d	23
Creating a <i>Partially</i> Mixed Event [FIXME IMAGES]	24
Incorporating one of the d - d correlations	27

A Case Study on $d-d-d$	27
Novel Estimation of the Nonresonant Background	28
A Case Study on $d-d-d$	28
Statistical Check	29
Another Case Study: $p-t-\alpha$ (No Signal)	30
Another Case Study: $p-t-\alpha$ (Crude Double Gaussian)	31
$7\alpha E^*$ spectrum (no signal model)	32
$7\alpha E^*$ spectrum (no signal model; hit pattern corrected)	33
Conclusions	34
Outline	35